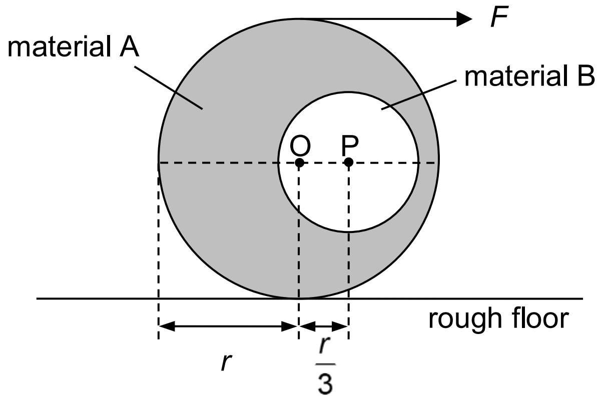
$a = (3000 – 800) / 1000 = 2.2$ m s^{-2}

5 A cylinder of radius *r*, made of material A and B, is placed on a rough floor as shown. The portion made of material B has a radius of 0.45*r* and a density of ρ that is half that of material A. O is the centre of the cylinder and P is the centre of the portion made of material B. The distance between O and P is $\frac{r}{3}$. If the cylinder is entirely made of material A, its weight will be 90 N.

A force *F* is applied horizontally to the top of the cylinder so that O and P are at the same height from the floor as shown.



What is the force *F* required to keep the cylinder in this equilibrium position?

A 1.5 N **B** 2.0 N **C** 4.5 N **D** 7.5 N

Ans: A
If the entire cylinder is A, the system would be in equilibrium without *F*. With B, and the system requires *F* to be in equilibrium. Hence, the moment due to *F* (and the resulting friction) should be equal to that due to the weight of (portion of A that is replaced by B - B).
With *F* in place, translational equilibrium in the horizontal direction requires a friction *f* equal in magnitude to *F* but acting in the opposite direction to *F*. *F* and *f* form a couple.
Weight of section replaced by (B) = $W_A - W_B = (\rho_A - \rho_B)(V_B)\,g = ½\,\rho_A V_B\,g$
Taking centre of mass of cylinder A as pivot
Moment due to the weight of section replaced by B
$= ½\,\rho_A\,\pi(0.45r)^2\,\ell g \times r/3 = 0.03375r^3\,\pi\,\rho_A\,lg$
Since $\rho_A V_A\,g = \rho_A\,\pi(r)^2\,\ell g = 90$ N,
$0.03375r^3\,\pi\,lg = 3.04\,r$ N
Moment due to the couple of *F* and *f* = *F* × 2*r*
Equating the two moments, *F* = 1.5 N

6 Two identical uniform rods, each of weight *W*, are hinged together to form a structure which is